x^2 +1 is continuous on [-1,3]

Yes — $f(x) = x^2 + 1$ is continuous on $[-1, 3]$.

Reason (in standard calculus terms):

- x^2 is a polynomial, and **all polynomials are continuous for every real** x.
- The constant function 1 is continuous everywhere.
- The sum of continuous functions is continuous.

Therefore, $x^2 + 1$ is continuous on all $\mathbb{R}$, so in particular it is continuous on the closed interval $[-1, 3]$.

should we check through limit and f(a) exist or not

You *can*, but for $x^2 + 1$ it's not necessary because it's a polynomial.

When you should check by limits

Use the "limit equals function value" test when:

- the function is **piecewise**, involves $|x|$, $\frac{1}{x}$, roots, etc.
- you suspect a **break** (hole, jump, vertical asymptote)
- you're checking continuity at a **specific point** a

The formal test (at a point a)

A function is continuous at a if all three hold:

1. $f(a)$ exists
2. $\lim_{x \to a} f(x)$ exists (left = right)
3. $\lim_{x \to a} f(x) = f(a)$

For $f(x) = x^2 + 1$:

- $f(a) = a^2 + 1$ exists for every real a
- $\lim_{x \to a}(x^2 + 1) = a^2 + 1$ (because polynomials have limits equal to substitution)
- so $\lim_{x \to a} f(x) = f(a)$

On an interval $[-1, 3]$

- For interior points $(-1, 3)$, continuity is as above.
- At endpoints, you use one-sided limits:
  - continuous at -1 if $\lim_{x \to -1^+} f(x) = f(-1)$
  - continuous at 3 if $\lim_{x \to 3^-} f(x) = f(3)$

It will satisfy these automatically.

So: **Checking via limits is the definition**, but for a polynomial you can directly state: "polynomials are continuous everywhere," and you're done.

Checking via limits is the definition step by step

Sure. Let $f(x) = x^2 + 1$. I'll apply the **definition of continuity using limits**, step by step.

Goal

Show f is continuous on $[-1, 3]$.
That means:

- For every a in $(-1, 3)$: $\lim_{x \to a} f(x) = f(a)$
- At the endpoints:
  - $\lim_{x \to -1^+} f(x) = f(-1)$
  - $\lim_{x \to 3^-} f(x) = f(3)$

A) Continuity at an interior point $a \in (-1, 3)$

Step 1: Check that $f(a)$ exists

$$f(a) = a^2 + 1$$

This is a real number for any real a, so $f(a)$ exists.

Step 2: Find $\lim_{x \to a} f(x)$

$$\lim_{x \to a}(x^2 + 1)$$

Use limit laws:

$$\lim_{x \to a}(x^2 + 1) = \lim_{x \to a} x^2 + \lim_{x \to a} 1$$

Now evaluate each:

- $\lim_{x \to a} 1 = 1$
- $\lim_{x \to a} x^2 = a^2$ (because squaring is continuous / substitution works for polynomials)

So:

$$\lim_{x \to a}(x^2 + 1) = a^2 + 1$$

Step 3: Compare $\lim_{x \to a} f(x)$ with $f(a)$

$$\lim_{x \to a} f(x) = a^2 + 1 = f(a)$$

So f is continuous at every interior point $a \in (-1, 3)$.

B) Continuity at the endpoint -1 (right-hand limit)

Step 1: Compute $f(-1)$

$$f(-1) = (-1)^2 + 1 = 1 + 1 = 2$$

Step 2: Compute the right-hand limit

$$\lim_{x \to -1^+}(x^2 + 1) = (-1)^2 + 1 = 2$$

Step 3: Compare

$$\lim_{x \to -1^+} f(x) = 2 = f(-1)$$

So f is continuous at -1 on the interval.

C) Continuity at the endpoint 3 (left-hand limit)

Step 1: Compute $f(3)$

$$f(3) = 3^2 + 1 = 9 + 1 = 10$$

Step 2: Compute the left-hand limit

$$\lim_{x \to 3}(x^2 + 1) = 3^2 + 1 = 10$$

Step 3: Compare

$$\lim_{x \to 3} f(x) = 10 = f(3)$$

So f is continuous at 3 on the interval.

Conclusion

- Continuous at every $a \in (-1, 3)$
- Continuous at -1 using right-hand limit
- Continuous at 3 using left-hand limit

Therefore, $f(x) = x^2 + 1$ is **continuous on** $[-1, 3]$.

If you want, I can also show the left-hand limit and right-hand limit explicitly equal (LHL = RHL) for a general a (though it's the same here).

but here LHL and RHL is not equal then how limit exist

For a **two-sided** limit $\lim_{x \to a} f(x)$ to exist, you are correct about the rule:

$$\lim_{x \to a} f(x) \text{ exists } \iff \lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x).$$

So if **LHL** $\neq$ **RHL**, then the **two-sided** limit does **not** exist.

But here is the key point for your question:

1) For $f(x) = x^2 + 1$, LHL and RHL are actually equal (for any interior point)

Take any point a (like $a = 0$, $a = 2$, etc.). Then:

Right-hand limit

$$\lim_{x \to a^+}(x^2 + 1) = a^2 + 1$$

Left-hand limit

$$\lim_{x \to a^-}(x^2 + 1) = a^2 + 1$$

Because whether x approaches a from the left or right, x^2 approaches a^2. So:

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = a^2 + 1$$

Therefore the two-sided limit exists.

If you tell me the exact point a where you think they're not equal, I can compute both sides with you and show it.

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2) At endpoints of an interval, we do not require both LHL and RHL

When we say "continuous on $[-1, 3]$", the endpoint checks are **one-sided**:

- At -1: only **right-hand** continuity matters (because the interval doesn't include values less than -1):

$$\lim_{x \to -1^+} f(x) = f(-1)$$

- At 3: only **left-hand** continuity matters:

$$\lim_{x \to 3^-} f(x) = f(3)$$

So even if you were thinking "LHL at -1"—that's not required **for continuity on the interval** $[-1, 3]$.

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3) Why you might think LHL and RHL "aren't equal"

Common confusion:

- **At** -1, the "left side" is outside the interval, so you may think "LHL doesn't match." But continuity on $[-1, 3]$ uses $\lim_{x \to -1^+}$, not a full two-sided limit.
- For $x^2 + 1$, even the full two-sided limit at -1 exists in $\mathbb{R}$ anyway, and equals 2.

If you reply with the point a where you believe LHL $\neq$ RHL (for example "at -1" or "at 0"), I'll compute both one-sided limits explicitly, step-by-step, and we'll pinpoint the exact misunderstanding.